A Data Warehouse Approach to understanding EV Adoption in Washington State

SUBMITTED BY

GOPI DODDA

TABLE OF CONTENTS

1. Executive Summary..... 3
2. Problem Statement..... 3
3. Literature Review..... 4
4. Data Collection and Preparation..... 4
5. Database Design..... 5
6. Entity Relationship Diagram..... 8
7. Exploratory Data Analysis..... 8
8. Reporting..... 12
9. Storytelling..... 15
10. Conclusion..... 19
11. References..... 20

EXECUTIVE SUMMARY

Electric vehicle (EV) registrations in Washington State provide a valuable opportunity to analyze how vehicle characteristics, income levels, geographic regions and policy classifications relate within a unified analytical environment. However, the raw datasets containing this information are stored separately and follow different formats, which limits their usefulness for multi-dimensional analysis.

This project develops a data warehouse designed to integrate EV population records, county-level median income data, and clean-fuel tax-exemption classifications into a single analytical structure. The warehouse is built using a STAR schema to support efficient querying and trend evaluation across time, vehicle attributes, counties and policy-related dimensions.

The project includes data collection, cleaning and transformation, dimensional modeling, ETL development in Oracle SQL Developer and exploratory SQL analysis. By consolidating the data into a structured warehouse, the project enables systematic examination of EV registration patterns over the last decade and provides a foundation for reporting, visualization, and further modeling activities.

PROBLEM STATEMENT

Electric vehicle population data and related socioeconomic datasets in Washington State exist in separate, heterogeneous formats. When used individually, these datasets provide only a partial view of EV registration patterns. To explore how registrations differ by geography, vehicle type, model year, and income levels – and to identify the presence of tax-exempt vehicles – there is a need for a unified analytical structure.

This project addresses the challenge of integrating EV registrations, county-level income values, and tax-exemption classifications into a dimensional data warehouse designed for analytical workloads. The focus is on enabling a specific analytical scenario: **examining EV registration patterns over the last 15 years and understanding how these patterns distribute across counties, income levels, manufacturers, and tax-exemption groups**.

The significance of this challenge lies in the ability to perform multi-dimensional analysis efficiently, observe trends across time and geography, and prepare data that can support reporting, visualization, or subsequent modeling tasks.

LITERATURE REVIEW

Multiple publicly available studies and transportation analytics reports were reviewed to contextualize the data elements used in this project. Several recurring themes appear in EV-related literature:

1. **Electric Vehicle Population Tracking**
   Many state agencies maintain EV population dashboards to monitor growth trends and identify regional differences in adoption.

2. **Income and Demographic Context**
   Income data is often incorporated in transportation studies when examining vehicle choice patterns or evaluating infrastructure needs.

3. **Clean-Fuel Incentives**
   Tax credits, exemptions and other clean-fuel incentives are frequently analyzed to understand how they align with vehicle registration data.

4. **Data Warehousing Concepts**
   Tutorials and reference materials on dimensional modeling, surrogate keys, ETL workflows, and STAR schemas guided the database design.

These sources collectively influenced the selection of dimensions (time, county, vehicle, vehicle type, income, tax-exemption) and shaped the analytical focus of the warehouse.

DATA COLLECTION AND PREPARATION

Data Sources

Three primary datasets were used:

1. **EV Population Data**
   Contains registration records for battery-electric and plug-in hybrid vehicles in Washington State, including make, model, model year, county, electric range, and MSRP.

2. **Median Household Income Dataset**
   Provides median household income by county for all years from 2000–2024.

3. **Tax Exemption Dataset**
   Includes vehicles classified as eligible for clean-fuel tax exemptions, identified by alternative-fuel type, model description, model year, and make.

Data Cleaning & Transformation

The following preparation steps were carried out:

- Removed non-essential, high-cardinality fields such as Vehicle Identification Number (VIN), city, state, postal code, and legislative district.
- Standardized text fields using UPPER() and TRIM().
- Filtered all datasets to model years 2010-2024 to maintain consistent temporal coverage and limit storage usage.
- Reshaped income data from a pivoted format to a row-oriented structure.
- Normalized vehicle model names to enhance matching accuracy with the tax-exemption dataset.
- Ensured alignment of county names, model years, and vehicle attributes.
- Created surrogate keys for each dimension table.
- Implemented initial ETL processes to load the cleansed data into Oracle SQL Developer.

These preparation steps produced coherent and joinable datasets suitable for dimensional modeling.

DATABASE DESIGN

The database design includes both a transactional (OLTP) model and an analytical (OLAP) dimensional model, demonstrating the difference between operational and analytical processing.

Transactional Model (OLTP)

The OLTP design represents a simplified structure of how EV registration data might be captured operationally.

Typical entities include:

- REGISTRATION – registration date, vehicle, county
- VEHICLE – make, model, model year, base attributes
- OWNER – owner details and county of residence
- COUNTY – list of counties

This schema represents row-level transactions and supports insert/update operations rather than analytical queries.

Dimensional Model (OLAP / STAR Schema)

The STAR schema was designed to support multi-dimensional analysis across time, geography, vehicle characteristics, income, and tax-exemption categories.

Dimension Table Descriptions

1. **TIME_DIM**

   Stores the temporal information for EV registrations to support year-based aggregations.
   Primary Key: TIME_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. TIME_ID

2. **VEHICLE_DIM**

   Describes vehicle make, model, and model year to support manufacturer and model-level analysis.
   Primary Key: VEHICLE_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. VEHICLE_ID

3. **VEHICLE_TYPE_DIM**

   Categorizes vehicles by EV type and Clean Alternative Fuel Vehicle (CAFV) eligibility.
   Primary Key: VEHICLE_TYPE_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. VEHICLE_TYPE_ID

4. **COUNTY_DIM**

   Represents the counties associated with vehicle registration.

   Primary Key: COUNTY_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. COUNTY_ID

5. **INCOME_DIM**

   Stores income values by county and year to support socioeconomic comparisons.

   Primary Key: INCOME_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. INCOME_ID

6. **TAX_EXEMPTION_DIM**

   Contains the clean-fuel tax-exemption classifications for eligible vehicle models.

   Primary Key: TAX_ID
   Foreign Key in Fact table: FACT_EV_REGISTRATION. TAX_ID

Fact Table Description

- **FACT_EV_REGISTRATION**

  Contains one record per EV registration and links to all dimensions.

  Primary Key: FACT_ID
  Foreign Keys: TIME_ID, VEHICLE_ID, VEHICLE_TYPE_ID, COUNTY_ID, INCOME_ID, TAX_ID
  Measures: EV_COUNT, ELECTRIC_RANGE, BASE_MSRP

- To support analytical querying, the dimensional model was structured so that the FACT_EV_REGISTRATION table is fully constructed from the surrogate keys of the surrounding dimension tables. Each record in the fact table references the appropriate entries in TIME_DIM, VEHICLE_DIM, VEHICLE_TYPE_DIM, COUNTY_DIM, INCOME_DIM, and TAX_EXEMPTION_DIM, creating a central point of integration for all relevant attributes and measures.

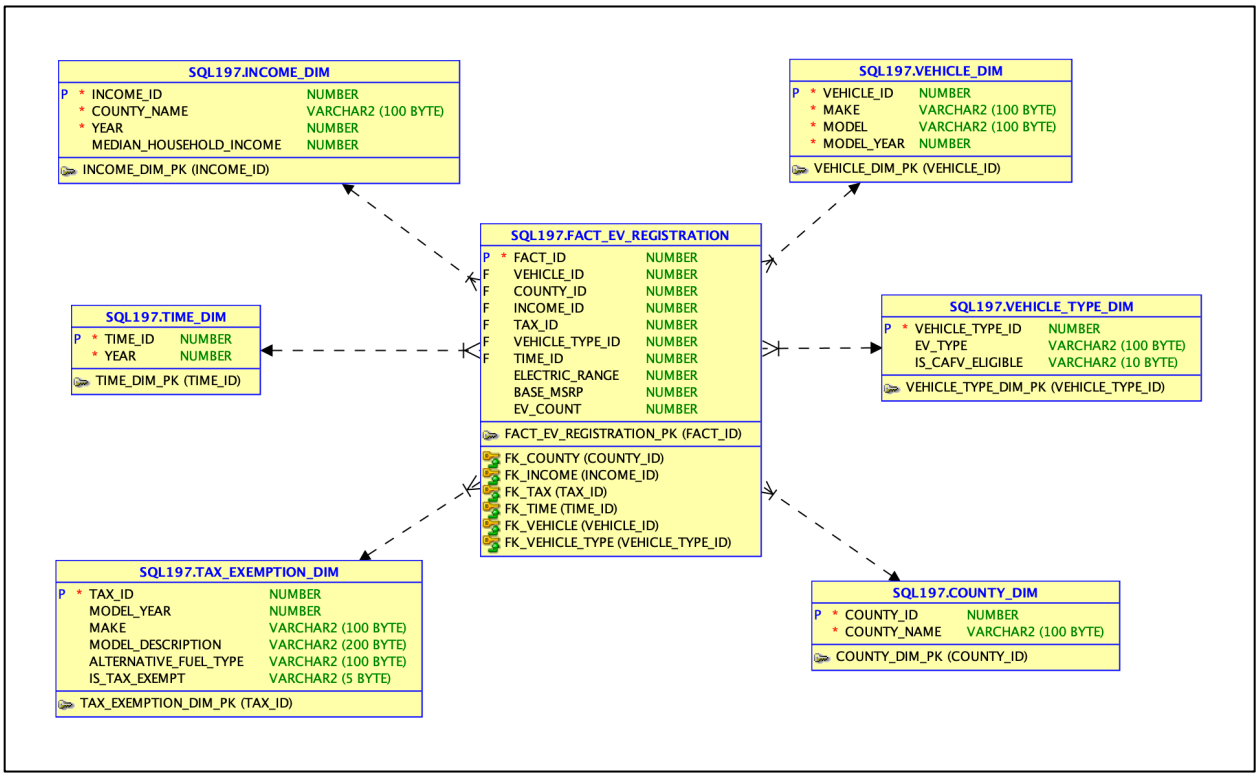
Data Cubes

The STAR schema supports cubes along dimensions such as:

- Time (year)
- Geography (county)
- Vehicle (make, model, model year)
- EV Type
- Income
- Tax Exemption category

These cubes enable slicing, dicing, roll-ups, and drilldowns across all analytical perspectives.

ENTITY RELATIONSHIP DIAGRAM



EXPLORATORY DATA ANALYSIS

Exploratory analysis was conducted to understand the structure and patterns of the integrated dataset. SQL queries were executed across the STAR schema to produce summary tables and descriptive statistics.

1. EV Registrations by Year

Objective: To observe how the number of EV registrations changes annually within the dataset scope (2010-2024).

SQL Query:

```
SELECT year, SUM(ev_count) AS total_registrations
FROM FACT_EV_REGISTRATION a
JOIN TIME_DIM b ON a.time_id = b.time_id
GROUP BY year
ORDER BY year DESC;
```

Output:

	YEAR	TOTAL_REGISTRATIONS
1	2024	32688
2	2023	35628
3	2022	15536
4	2021	10048
5	2020	9194
6	2019	7276
7	2018	7678
8	2017	3358
9	2016	3656
10	2015	2022
11	2014	866
12	2013	804
13	2012	218
14	2011	130

Summary:
EV registrations have increased steadily over the last decade, with especially rapid growth occurring after 2018. The years 2022–2024 show the largest annual increases, indicating accelerating adoption in recent years. This trend suggests expanding consumer confidence, maturing technology, and strong policy support for EV adoption across the state.

2. EV Registrations by County

Objective: To compare registration counts across Washington State counties and identify regions with higher EV presence.

SQL Query:

```
SELECT county_name, SUM(ev_count) AS total_registrations
FROM FACT_EV_REGISTRATION a
JOIN COUNTY_DIM b ON a.county_id = b.county_id
GROUP BY county_name
ORDER BY total_registrations DESC;
```

Output:

	COUNTY_NAME	TOTAL_REGISTRATIONS
1	King	74704
2	Snohomish	14622
3	Clark	9350
4	Pierce	6934
5	Kitsap	4394
6	Thurston	4028
7	Spokane	2218
8	Whatcom	2052
9	Yakima	1194
10	Island	1088

Summary:
King County accounts for the majority of registrations, followed by Snohomish and Clark counties. These counties contain major population centers, reflecting the strong link between EV adoption, urbanization, and infrastructure availability.

3. EV Registrations by Vehicle Make

Objective: To understand the composition of EV manufacturers in the dataset and identify leading brands.

SQL Query:

```
SELECT make, SUM(ev_count) AS total_registrations
FROM FACT_EV_REGISTRATION a
JOIN VEHICLE_DIM b ON a.vehicle_id = b.vehicle_id
GROUP BY make
ORDER BY total_registrations DESC;
```

Output:

	MAKE	TOTAL_REGISTRATIONS
1	TESLA	74324
2	RIVIAN	9064
3	FORD	8462
4	VOLKSWAGEN	5750
5	BMW	4938
6	KIA	4712
7	NISSAN	4606
8	CHEVROLET	3606
9	HYUNDAI	2860
10	AUDI	2138

Summary:
Tesla registrations far exceed those of any other manufacturer, followed by Rivian, Ford, Volkswagen, BMW, Kia, Nissan, and Chevrolet. This dominance points to strong brand preference, early market entry and Tesla's extensive charging ecosystem, which continues to influence consumer choices.

4. EV Registration Distribution Across Income Categories

Objective: To group county-level median household income into categorical income slabs and analyze how EV registrations are distributed across different income categories.

SQL Query:

```sql
WITH
cte AS (
    SELECT a.*,
      CASE
         WHEN median_household_income < 60000 THEN 'Low Income'
         WHEN median_household_income BETWEEN 60000 AND 74999 THEN 'Lower-Middle Income'
         WHEN median_household_income BETWEEN 75000 AND 89999 THEN 'Middle Income'
         WHEN median_household_income BETWEEN 90000 AND 109999 THEN 'Upper-Middle Income'
         ELSE 'High Income'
      END AS income_category
    FROM FACT_EV_REGISTRATION a
    LEFT JOIN INCOME_DIM b ON a.income_id = b.income_id
)

SELECT income_category, SUM(ev_count) AS total_ev_count
FROM cte
GROUP BY income_category;
```

Output:

	INCOME_CATEGORY	TOTAL_EV_COUNT
1	High Income	59344
2	Upper-Middle Income	36854
3	Middle Income	18022
4	Lower-Middle Income	11670
5	Low Income	3212

Summary:
EV registrations rise significantly with income, with high and upper-middle income categories accounting for the majority of registrations. This pattern indicates that affordability, home charging capability, and access to incentives play a critical role in shaping EV adoption across socioeconomic groups.

5. **Distribution of EV Types**

Objective: To compare counts of Battery EVs and Plug-in Hybrid EVs.

SQL Query:

```
SELECT ev_type, SUM(ev_count) AS total_registrations
FROM FACT_EV_REGISTRATION a
JOIN VEHICLE_TYPE_DIM b ON a.vehicle_type_id = b.vehicle_type_id
GROUP BY ev_type
ORDER BY total_registrations DESC;
```

Output:

	EV_TYPE	TOTAL_REGISTRATIONS
1	Battery Electric Vehicle (BEV)	124594
2	Plug-in Hybrid Electric Vehicle (PHEV)	4508

Summary:
Battery EVs (BEVs) make up over 95% of total registrations, while Plug-in Hybrids (PHEVs) account for only a small fraction. The overwhelming preference for BEVs highlights Washington's readiness for full electrification and suggests increasing confidence in EV range, charging infrastructure, and long-term sustainability benefits.

REPORTING

This section applies advanced analytical SQL functions to extract deeper insights from the EV registration data. Functions such as LAG(), CORR(), NTILE(), and window functions enable trend analysis, ranking, correlation assessment, and richer reporting beyond what was covered during exploratory analysis. Each query below demonstrates a specific analytical technique and highlights meaningful aspects of EV adoption across Washington State.

1. **Year-over-Year Change in EV Registrations (using LAG)**

Objective: To measure year-over-year growth by comparing each year's registrations to the previous year, enabling identification of periods with the strongest or weakest growth.

SQL Query:

```
SELECT year, SUM(ev_count) AS total_registrations, LAG(SUM(ev_count)) OVER (ORDER BY year) AS prev_year_registrations,
    ROUND((SUM(ev_count) - LAG(SUM(ev_count)) OVER (ORDER BY year)) / (SUM(ev_count)) * 100.0,2) AS percent_yoy_change
FROM FACT_EV_REGISTRATION a
JOIN TIME_DIM b ON a.time_id = b.time_id
GROUP BY year
ORDER BY year;
```

Output:

	YEAR	TOTAL_REGISTRATIONS	PREV_YEAR_REGISTRATIONS	PERCENT_YOY_CHANGE
1	2011	130	(null)	(null)
2	2012	218	130	40.37
3	2013	804	218	72.89
4	2014	866	804	7.16
5	2015	2022	866	57.17
6	2016	3656	2022	44.69
7	2017	3358	3656	-8.87
8	2018	7678	3358	56.26
9	2019	7276	7678	-5.53
10	2020	9194	7276	20.86
11	2021	10048	9194	8.5
12	2022	15536	10048	35.32
13	2023	35628	15536	56.39
14	2024	32688	35628	-8.99

Summary:
The year-over-year percentages show strong EV adoption growth from 2011 through 2016, followed by periodic fluctuations between 2017 and 2020. Growth accelerates again sharply after 2021, with the largest jumps in 2022 and 2023, indicating renewed momentum in statewide EV adoption. Occasional negative values reflect short-term dips but do not disrupt the long-term upward trend.

2. Correlation Between County Income and EV Registrations (Using CORR)

Objective: To measure the strength of the relationship between county median income and EV adoption levels

SQL Query:

```sql
SELECT ROUND(CORR(ev_total, median_income),2) AS ev_income_correlation
FROM (
    SELECT county_name,
        AVG(median_household_income) AS median_income,
        SUM(ev_count) AS ev_total
    FROM FACT_EV_REGISTRATION a
    JOIN INCOME_DIM b ON a.income_id = b.income_id
    GROUP BY county_name
);
```

Output:

	EV_INCOME_CORRELATION
1	0.68

Summary:
The correlation coefficient of 0.68 indicates a strong positive relationship between median household income and EV registrations. This suggests that higher-income counties tend to adopt EVs at significantly greater rates, reinforcing the income-based adoption gap observed earlier in the EDA.

3. EV Adoption Quartiles by County (Using NTILE)

Objective: To determine how counties rank in terms of total EV registrations, grouping them into four adoption levels.

SQL Query:

```
SELECT county_name, SUM(ev_count) AS total_registrations,
    NTILE(4) OVER (ORDER BY SUM(ev_count) DESC) AS adoption_quartile
FROM FACT_EV_REGISTRATION a
JOIN COUNTY_DIM b ON a.county_id = b.county_id
GROUP BY county_name
ORDER BY total_registrations DESC;
```

Output:

	COUNTY_NAME	TOTAL_REGISTRATIONS	ADOPTION_QUARTILE
1	King	74704	1
2	Snohomish	14622	1
3	Clark	9350	1
4	Pierce	6934	1
5	Kitsap	4394	1
6	Thurston	4028	1
7	Spokane	2218	1
8	Whatcom	2052	1
9	Yakima	1194	1
10	Island	1088	1
11	Skagit	1008	2
12	Benton	994	2
13	Cowlitz	966	2
14	Jefferson	598	2
15	Lewis	570	2

Summary:
The quartile analysis shows that the highest-adoption counties, such as King, Snohomish, Clark, and Pierce, dominate the top quartile. These counties have far higher EV counts compared to counties in the lower quartiles, highlighting geographic concentration and uneven adoption across the state. The results reveal clear disparities between urban and rural regions.

4. Market Share of Manufacturers (Using Window SUM)

Objective: To measure each manufacturer's contribution to the total EV market.

SQL Query:

```
SELECT make, SUM(ev_count) AS total_ev_count,
    ROUND( SUM(ev_count) * 100.0 / SUM(SUM(ev_count)) OVER (), 2) AS market_share_percentage
FROM FACT_EV_REGISTRATION a
JOIN VEHICLE_DIM b ON a.vehicle_id = b.vehicle_id
GROUP BY make
ORDER BY total_ev_count DESC;
```

Output:

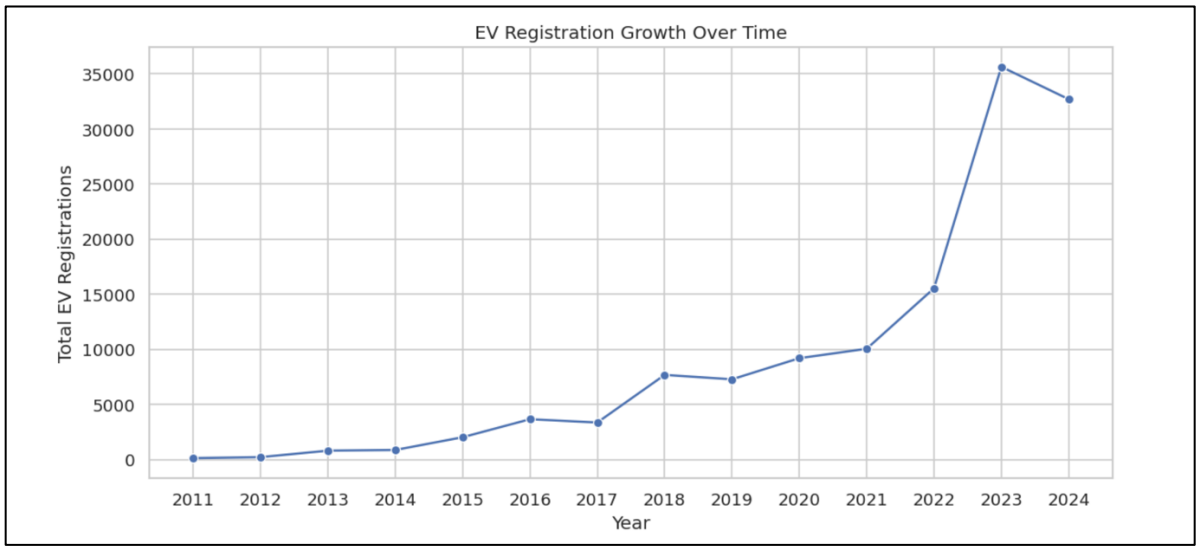
	MAKE	TOTAL_EV_COUNT	MARKET_SHARE_PERCENTAGE
1	TESLA	74324	57.57
2	RIVIAN	9064	7.02
3	FORD	8462	6.55
4	VOLKSWAGEN	5750	4.45
5	BMW	4938	3.82
6	KIA	4712	3.65
7	NISSAN	4606	3.57
8	CHEVROLET	3606	2.79
9	HYUNDAI	2860	2.22
10	AUDI	2138	1.66

Summary:
Tesla accounts for over 57% of all EV registrations, far surpassing every other manufacturer. Rivian, Ford, Volkswagen, and BMW follow but hold comparatively small shares, illustrating a highly concentrated market. The distribution indicates strong consumer preference for Tesla vehicles and limited competition among other manufacturers.

STORYTELLING

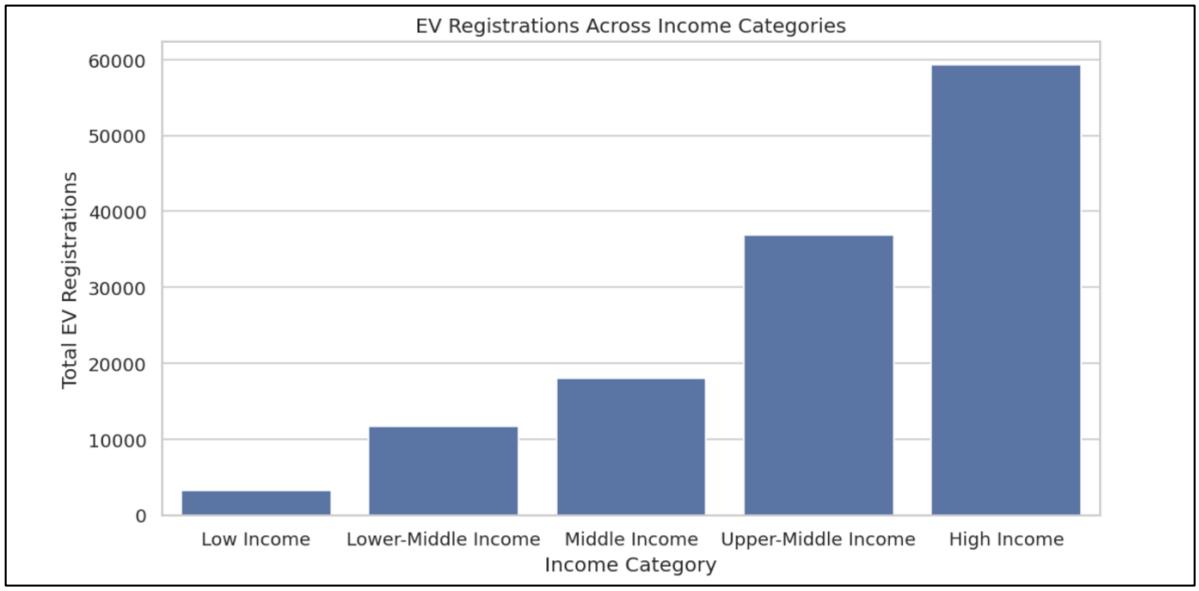
This section presents a narrative interpretation of key visualizations created from the analytical dataset. The aim is to translate quantitative findings into meaningful insights that explain how electric vehicle (EV) adoption varies across time, geography, income groups, and manufacturers within Washington State. Together, these visualizations provide a clear picture of the evolving EV landscape and the factors influencing statewide adoption patterns.

1. EV Registration Growth Over Time



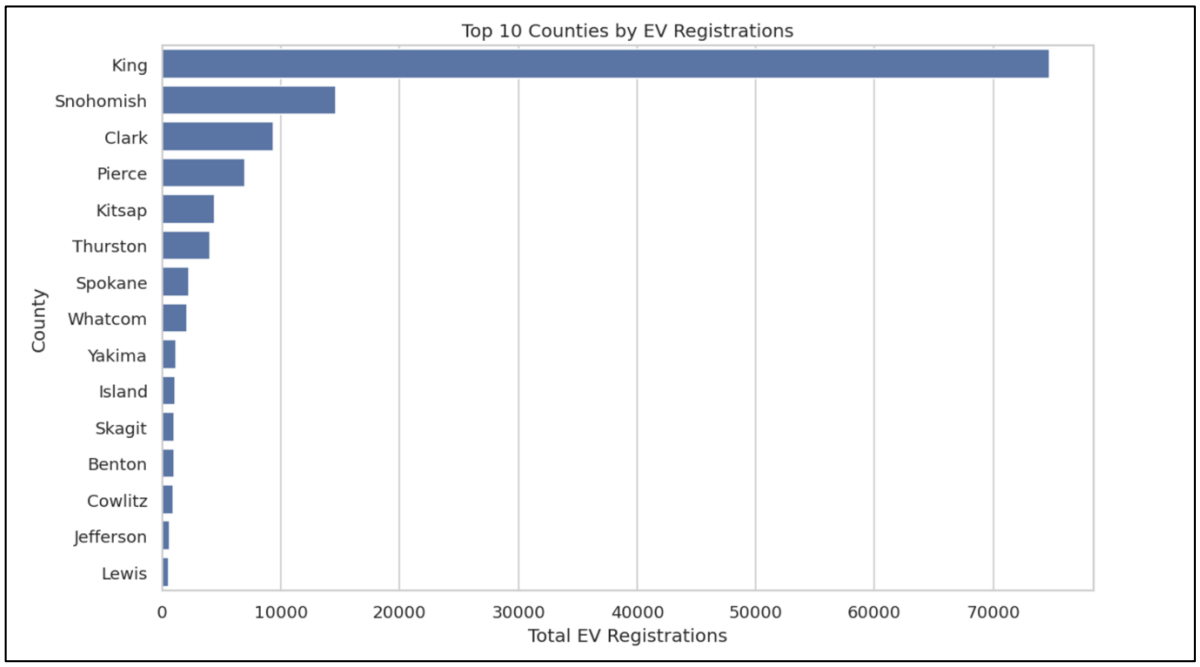
The time-series visualization highlights a strong upward trend in EV registrations from 2011 through 2024. Early adoption grew steadily from 2011 to 2017, but EV uptake accelerated significantly beginning around 2018. The most dramatic increase appears between 2022 and 2023, reflecting broader model availability, improved battery technology, and growing consumer confidence. Although 2024 shows a slight decline from the 2023 peak, the overall trend indicates that EV adoption in Washington State is expanding rapidly and consistently.

2. EV Registrations Across Income Categories



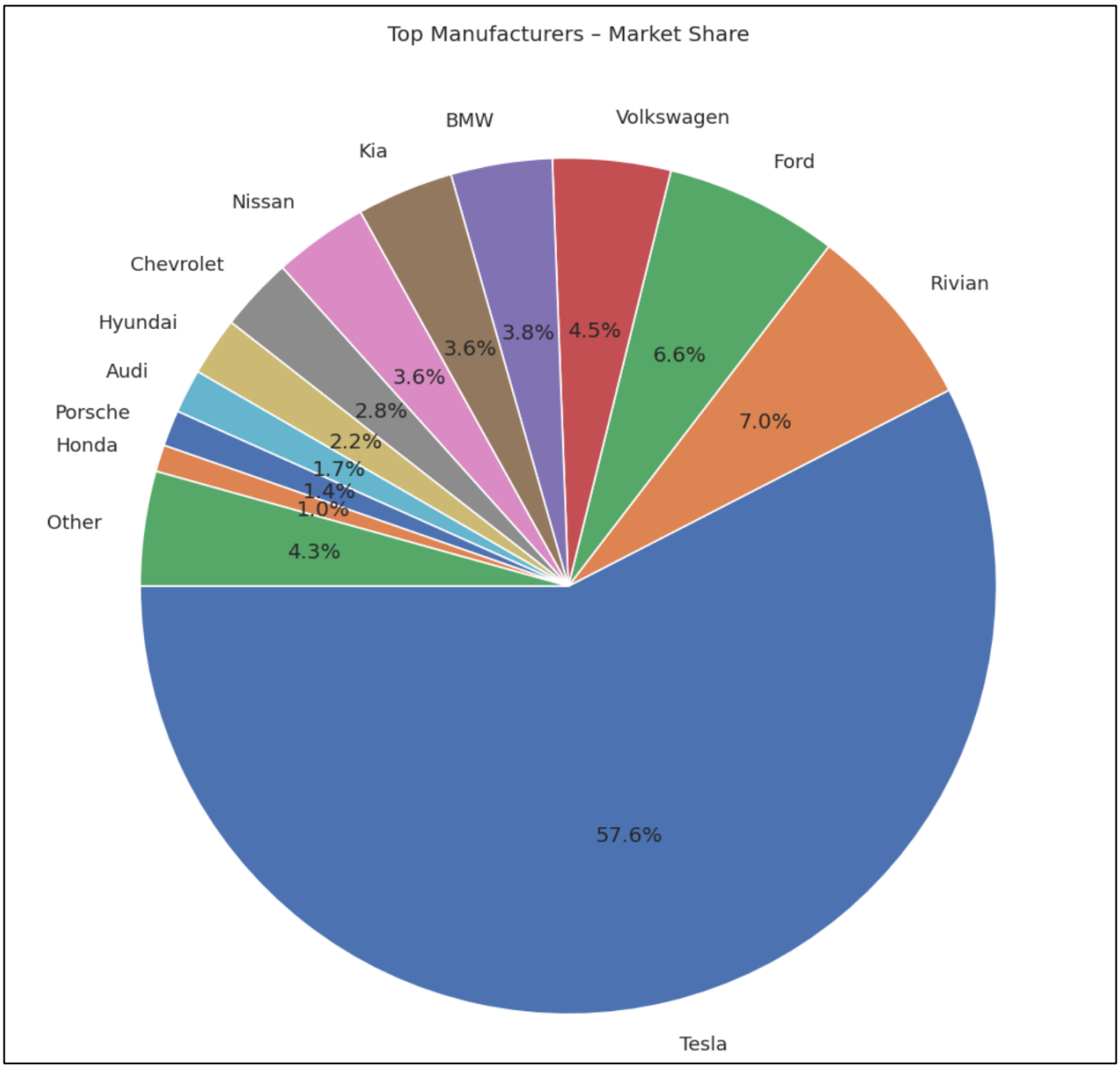
The income-based visualization reveals a clear association between median household income and EV adoption. High-income and upper-middle-income groups account for the largest share of registrations by a wide margin. Middle-income categories show moderate adoption, while lower-income groups lag considerably behind. This pattern suggests that affordability and access remain critical factors shaping EV adoption. Higher-income households likely benefit from greater purchasing power, easier access to home charging, and stronger alignment with EV market pricing, reinforcing disparities across demographic groups.

3. Top 15 Counties by EV Registrations



The geographic distribution of EV registrations shows that adoption is heavily concentrated in a small number of counties. King County stands out dramatically, with more than 70,000 registrations – far exceeding all others. Snohomish, Clark, and Pierce Counties follow with strong adoption, while the remaining counties in the top 10 show comparatively smaller but still meaningful uptake. This concentration reflects differences in population density, economic conditions, urban infrastructure, and charging availability. The data suggests that EV adoption flourishes most in regions with strong economic and infrastructural support.

4. Market Share of EV Manufacturers



The market share visualization demonstrates a highly concentrated EV market. Tesla overwhelmingly dominates with over 57% of all registrations, highlighting its role as the primary driver of EV adoption in the state. Rivian, Ford, Volkswagen, BMW, Kia, and Nissan account for smaller but notable shares, while the remaining manufacturers collectively form a small "Other" category. Tesla's leadership reflects its early market presence, strong charging ecosystem, and broad consumer appeal, while the diversity among other manufacturers suggests a gradually expanding competitive landscape.

Summary:

Across all visualizations, a consistent narrative emerges: EV adoption in Washington State is growing rapidly but unevenly. Higher-income households and urban counties show significantly higher adoption, while rural and lower-income regions lag behind. At the same time, manufacturer dynamics reveal a dominant market leader alongside a range of emerging competitors. These insights together illustrate the economic, technological, and geographic factors shaping EV transition across the state, and they highlight opportunities for more inclusive and equitable policy design.

CONCLUSION:

This project used a dimensional data warehouse and analytical visualizations to better understand EV adoption patterns in Washington State. The primary findings and contributions are summarized below:

Key Findings:

- EV adoption has grown rapidly, with the strongest surge occurring between 2022 and 2023.
- High-income communities show the highest EV adoption, indicating that cost and affordability remain major adoption barriers.
- EV registrations are geographically concentrated, with King County leading by a large margin compared to all other counties.
- Tesla holds a dominant market share, accounting for more than half of all EV registrations in the state.
- Counties with higher median household income tend to register more EVs, supporting a positive correlation between income and adoption.

Project Contributions:

- Designed and implemented a clean, efficient star schema integrating seven dimension and fact tables.
- Built a unified analytical dataset enabling fast aggregation, correlation analysis, and flexible visualization.
- Demonstrated the value of dimensional modeling for drilling down into time, geography, income, vehicle characteristics, and tax attributes.
- Produced a set of clear, data-driven visualizations to support reporting and storytelling.

Limitations:

- The dataset does not include charging infrastructure availability, which strongly influences EV adoption.
- EV price data (MSRP) contains many zeros, limiting its analytical usefulness.
- County-level income data may not capture within-county socioeconomic variation.
- Some model descriptions in the tax exemption dimension are inconsistent, which may reduce join accuracy.

Next Steps:

- Incorporate charging station density or infrastructure data to better explain geographic differences in adoption.
- Explore predictive modeling to forecast future EV growth or identify emerging adoption hotspots.
- Add more detailed demographic data to support equity-focused analysis.
- Expand manufacturer analysis by including model-level trends, battery range improvements, and price segmentation.

Overall, this project demonstrates how dimensional modeling, SQL analytics, and visualization techniques can work together to reveal clear patterns in EV adoption and support informed decision-making for sustainable transportation planning.

REFERENCES:

EV Data & Policy Sources

Washington State Department of Licensing. *Electric Vehicle Population Data*. Washington Open Data. https://data.wa.gov/

U.S. Census Bureau. *Small Area Income and Poverty Estimates (SAIPE)*. https://www.census.gov/programs-surveys/saipe.html

U.S. Department of Energy. *Alternative Fuels Data Center: Electric Vehicles*. https://afdc.energy.gov/

Dimensional Modeling & Data Warehousing

Inmon, W. H. (2005). *Building the Data Warehouse* (4th ed.). Wiley.

Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling* (3rd ed.). Wiley.